Keshav Sharma

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EDUCATION

Vellore Institute of Technology — **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) — *Graduation: May 2027*

SKILL SUMMARY

Languages: Python, SQL, Java, C/C++, JavaScript, R
Data Engineering and Database: ETL Pipelines, Data Collection, Data Processing, Data Validation, Data Quality, Feature Engineering, Data Warehousing Concepts, Data Modeling, MySQL, SQLite, Redis, SQL Query Optimization
Machine Learning & AI: Scikit-learn, TensorFlow, PyTorch, Time-Series Forecasting, Anomaly Detection, LLM Applications, Prompt Engineering, Generative AI, Model Evaluation
Cloud & Big Data:AWS (S3, Lambda, EC2 - Basic), Apache Airflow, PySpark, Databricks
Visualization: Power BI, Matplotlib, Tableau
Frameworks: FastAPI, Flask, React.js, Node.js
Developer Tools: Git, GitHub, Docker, GitHub Actions, REST APIs, CI/CD, PyTest

EXPERIENCE

VOIS – Vodafone Idea Foundation* |LINK** — ***Aug – Sep, 2025

- Processed and analyzed 8,500+ records from real-world Airbnb and Netflix datasets using an LLM-assisted analytics workflow, reducing manual analysis effort by 70%.
- Designed automated data preprocessing, validation, and exploratory analysis pipelines, generating 15+ business-ready insights with improved data quality and consistency.
- Leveraged Generative AI and prompt engineering to automate reporting workflows, reducing insight generation time by 60% while ensuring output reliability.

PROJECTS

***Customer Lifetime Value Engine* |LINK**

- Built an end-to-end ETL pipeline processing 1M+ customer transactions, performing data cleaning, validation, and feature engineering for predictive analytics.
- Developed BG/NBD and Gamma-Gamma models for customer lifetime value prediction, achieving an R^2 score of 0.87 on historical purchase data.
- Created interactive Power BI dashboards for customer segmentation and revenue forecasting, reducing reporting time by 80%.

***Real-Time Stock Market Anomaly Detector* |LINK**

- Developed a real-time data pipeline processing 2M+ NSE market records, integrating structured market data with financial news sentiment analysis.
- Built Isolation Forest and LSTM Autoencoder models achieving 88% precision and 85% recall for anomaly detection on validation datasets.
- Optimized preprocessing and inference workflows, reducing anomaly detection latency by 35% and enabling near real-time analytics.

***AI-Powered Expense Receipt Scanner & Analytics Platform* |LINK**

- Built a full-stack receipt processing platform that extracted structured data from 12,000+ receipts with approximately 96% OCR accuracy using EasyOCR and OpenCV.
- Developed secure FastAPI backend services with JWT authentication, Google OAuth, and REST APIs, reducing average API response time to under 180 ms.
- Automated expense categorization and analytics through NLP-based processing, reducing manual bookkeeping effort by 80% while providing interactive financial dashboards.

***RedisLite* |LINK**

- Engineered a Redis-inspired in-memory key-value database achieving 619K+ SET operations/sec, featuring a multithreaded TCP server, custom RESP parser, and O(1) LRU cache.
- Implemented Write-Ahead Logging (WAL with automatic crash recovery and validated the system using 46 automated PyTest test cases, ensuring high reliability and data integrity.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional |**LINK** 2026
- Prepare Data for ML APIs on Google Cloud Skill Badge |**LINK** 2026
- AWS APAC Solutions Architecture virtual experience program | **LINK** 2025